

Foundation of Blockchain Technology (IS501) Pre-Reading Material – Lecture 11

Blockchain Consensus Challenges & Solution - B

Estimated Reading Time: 15–20 minutes

Why This Topic Matters

Traditional blockchain consensus mechanisms such as Proof of Work require participants to perform computational work to reach agreement.

In a blockchain system combined with machine learning, it is desirable to use computational resources for **useful machine-learning tasks** rather than only cryptographic puzzles.

However, if blockchain participants compete based on machine-learning performance, the network must verify their claims.

This creates a challenge:

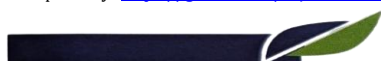
How can a participant prove that its model performs well without revealing the model?

Zero-Knowledge Proof of Training (ZKPoT) addresses this problem by using zero-knowledge proofs to verify training/model performance while protecting the underlying model.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the motivation for ZKPoT.
- Explain the limitations of traditional performance-based consensus.
- Describe the basic ZKPoT workflow.
- Explain model commitment.
- Explain the role of zk-SNARK in ZKPoT.
- Describe the role of IPFS.
- Explain transaction verification.



- Explain leader selection.
- Understand the role of two-thirds voting.
- Describe how a new block is generated.
- Compare ZKPoT with Proof of Work.

Activate Your Prior Knowledge

1. Proof of Work

In PoW, miners compete by solving computational puzzles.

Think: Is all this computation directly useful to an application such as machine learning?

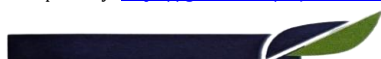
2. Machine-Learning Competition

Suppose five blockchain participants train models:

Participant Accuracy

A	91%
B	96%
C	89%
D	98%
E	93%

Think: Should Participant D automatically become the leader just because it claims 98% accuracy?



3. Model Privacy

Suppose Participant D is asked to provide its model so that everyone can verify the 98% accuracy.

Think: What privacy or security problems could occur?

Prerequisite Concepts

You should be familiar with:

- Blockchain
- Blocks and transactions
- Consensus mechanisms
- Proof of Work
- Federated Learning
- Machine-learning model training
- Zero-Knowledge Proofs
- zk-SNARK
- Cryptographic hashing
- IPFS

Core Concepts

1. From PoW to Learning-Based Consensus

Proof of Work requires miners to solve computational puzzles.

In learning-based blockchain systems, computational resources can instead be used for machine-learning tasks.

Examples include:

- Proof of Deep Learning (PoDL)
- Proof of Federated Learning (PoFL)
- Zero-Knowledge Proof of Training (ZKPoT)



The important idea is:

Useful computation can contribute to consensus.

2. The Problem with Performance-Based Consensus

Suppose:

Client A claims 95% accuracy.

The network needs to verify the claim.

One possible method is:

Obtain Model → Run Test Data → Calculate Accuracy

But this requires exposing the model.

Therefore:

Performance Verification → Model Disclosure → Privacy Risk

ZKPoT attempts to break this connection.

3. Basic ZKPoT Idea

The participant proves:

“My model achieves the claimed performance.”

without revealing:

- the complete model
- private model parameters
- private training information

The fundamental workflow is:

Model → ZKP → Proof

rather than:

Model → Other Nodes

4. ZKPoT Workflow

The overall process can be remembered as:

Task Publication



Global Model Distribution



Local Training



Model Quantization



Model Commitment



Accuracy Calculation



ZK Proof Generation



Transaction Submission





Proof Verification



Leader Selection



Model Aggregation



Block Generation

5. Local Training

Each participating client receives the global model.

The client trains the model using its own local dataset.

Therefore:

Local Data → Local Training → Local Model

The private training data does not need to be directly shared with the blockchain network.

6. Model Quantization

Machine-learning models commonly contain floating-point values.

For example:

0.73456

zk-SNARK computation requires values to be represented appropriately within a finite-field arithmetic framework.

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Therefore, the model may be converted into a suitable quantized representation.

The basic idea is:

Floating-Point Model



Quantization



Integer/Finite-Field Representation

This makes the model computation suitable for cryptographic proof generation.

7. Model Commitment

Before proving model performance, the client creates a **commitment** to the model.

Think of commitment as:

A sealed envelope containing the model.

Everyone can see that a commitment exists, but they cannot see the model inside it.

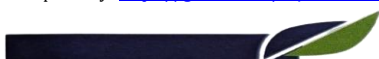
A commitment should provide:

Hiding

The commitment does not reveal the model.

Binding

The participant cannot change the committed model after creating the commitment.



8. Accuracy Calculation

The client evaluates its model on the specified test dataset.

For example:

Total test samples = 1000

Correct predictions = 950

Therefore:

Accuracy = $950 / 1000 = 95\%$

The client claims:

Accuracy = 95%

But the client still does not reveal the model.

9. ZK Proof Generation

Now the client generates a zero-knowledge proof.

The proof demonstrates that:

- the client possesses the committed model;
- the model is associated with the commitment;
- the model was evaluated correctly;
- the claimed accuracy is correct.

The important point is:

The proof reveals the validity of the computation, not the secret model itself.

10. Transaction Creation

The client creates a blockchain transaction containing relevant information such as:

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- Sender ID
- Task
- Claimed accuracy
- Model address
- Proof address
- Commitment
- Cryptographic information

The transaction becomes part of the blockchain's transaction pool.

11. Why IPFS?

Machine-learning models and cryptographic proofs can be large.

Putting all this information directly on-chain can increase:

- Blockchain storage
- Communication overhead
- Transaction size

Therefore, large objects can be stored in **IPFS**.

The blockchain stores the corresponding address/reference.

Conceptually:

Large Model/Proof



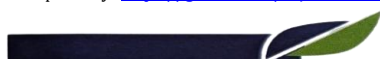
IPFS



IPFS Address



Blockchain



12. Transaction Verification

Suppose the transaction pool contains:

Client Claimed Accuracy

- A 91%
- B 97%
- C 94%
- D 88%
- E 96%

Transactions can be ordered according to claimed performance:

B → E → C → A → D

The blockchain nodes then verify the proofs.

The key principle is:

A high accuracy claim is not sufficient; the claim must have a valid proof.

13. Leader Selection

The candidate with the best verified performance can become the leader, subject to the consensus requirement.

The described ZKPoT protocol uses a **two-thirds voting threshold**.

If there are:

N = 12 participants

then:



Required votes = $2N/3$

= $2 \times 12 / 3$

= 8 votes

Therefore, a candidate receiving at least **8 valid votes** can satisfy the threshold.

14. Why Is Leader Selection Important?

The leader is responsible for important blockchain operations such as:

- coordinating the block-generation process;
- aggregating valid model updates;
- creating the new block;
- broadcasting the block to other nodes.

Therefore:

Leader selection → Consensus security

If a malicious or dishonest participant becomes leader, the integrity of the blockchain could be affected.

ZKPoT attempts to make leader selection dependent on **verifiable model performance**.

15. Model Aggregation

After the consensus process, valid model updates can be aggregated.

A common federated-learning approach is:

Federated Averaging (FedAvg)

Conceptually:

Local Model 1

-

Local Model 2

-

Local Model 3

...



Aggregation



Global Model

The updated global model can then be distributed for the next training round.

16. Block Generation

After leader selection and model aggregation, a new block is generated.

A simplified block can contain:

- Block number
- Timestamp
- Previous block hash
- Verified transactions
- Model information/reference
- Merkle root

The block is then broadcast to the blockchain network.



17. Complete Example

Suppose five clients submit the following claims:

Client Accuracy ZK Proof

A	95%	Valid
B	98%	Invalid
C	92%	Valid
D	96%	Valid
E	94%	Valid

The network first considers B because it claims the highest accuracy.

However:

B → 98% → Invalid Proof

Therefore B cannot be selected based only on its claim.

Next:

D → 96% → Valid Proof

If D receives the required consensus votes:

D → Leader

This demonstrates the central principle:

The highest claimed performance does not automatically determine the leader; the performance must be verifiable.

18. ZKPoT vs Proof of Work

Feature	Proof of Work	ZKPoT
Main computation	Cryptographic puzzle	ML training + proof
Primary objective	Blockchain consensus	Blockchain consensus + useful ML work
Energy demand	High	Avoids PoW puzzle mining
Model privacy	Not applicable	Important objective
Leader selection	Mining competition	Verified model performance
Verification	PoW verification	ZK proof verification
ML training	Not required	Central component

19. Important Trade-Off

ZKPoT does not mean that cryptographic proofs are computationally free.

Proof generation can require significant computational resources.

Therefore, there is a trade-off:

Privacy + Verifiability

versus

Proof Generation Cost

The important design objective is to obtain strong privacy and efficient verification while keeping proof-generation overhead manageable.



Guided Reading Questions

1. Why is Proof of Work computationally expensive?
2. What is learning-based consensus?
3. What problem occurs when blockchain nodes directly inspect a participant's model?
4. What is ZKPoT?
5. Why is model commitment required?
6. What is the role of zk-SNARK in ZKPoT?
7. Why is model quantization required?
8. Why is IPFS used?
9. What information is stored in a ZKPoT transaction?
10. How are transactions verified?
11. What is the purpose of the two-thirds voting threshold?
12. How is the leader selected?
13. What is the role of FedAvg?
14. How is the new block generated?
15. What is the major advantage of ZKPoT over traditional performance verification?

Reflection Activity

Write **100–150 words** explaining:

How can Zero-Knowledge Proof of Training improve the privacy of a blockchain-based federated-learning system?

Your answer should discuss:

- Blockchain
- Federated Learning
- Model privacy
- Zero-Knowledge Proof
- Consensus

Self-Assessment

Fill in the blanks:

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1. ZKPoT stands for Zero-Knowledge Proof of _____.
2. In ZKPoT, participants can prove model performance without revealing the _____.
3. A cryptographic commitment provides _____ and binding properties.
4. Large models and proofs can be stored using _____.
5. The blockchain stores references/addresses to data stored in _____.
6. Transactions can be sorted according to claimed model _____.
7. A candidate must pass proof _____ before becoming a leader.
8. The described protocol uses a _____ voting threshold.
9. The common federated-learning aggregation method used is _____.
10. The final result of the consensus process is recorded in a new blockchain _____.

Key Takeaway

ZKPoT combines three important technologies:

Blockchain

Provides:

Decentralization + Consensus + Immutability

Federated Learning

Provides:

Collaborative ML + Local Data Privacy

Zero-Knowledge Proof

Provides:

Verifiable Computation + Privacy

Together:

Federated Learning

↓

Zero-Knowledge Proof

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Verified Performance



Consensus



Leader Selection



Model Aggregation



Blockchain Block

Final Concept

ZKPoT allows blockchain participants to demonstrate that they have performed useful machine-learning work correctly, while avoiding the need to reveal the underlying model.

